SYSTEMS MANAGEMENT



DMTF PMCI Enhancements for OCP Hardware Management

Patrick Caporale *Distinguished Engineer, Lenovo*

Hemal Shah *Distinguished Engineer and Architect, Broadcom Inc.*

OCP PLATINUM Member

Disclaimer

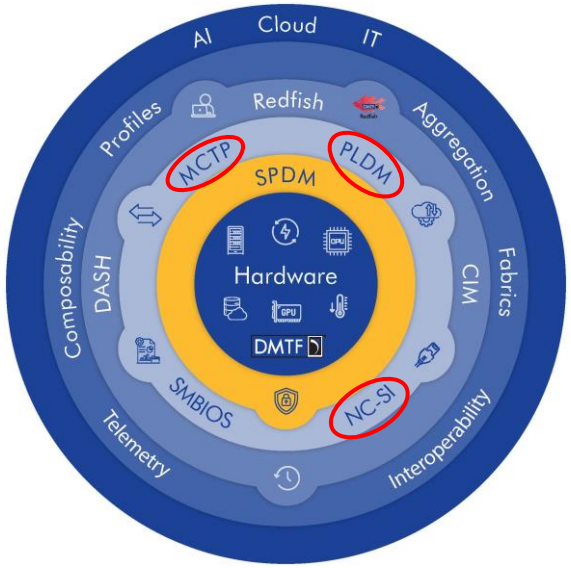
- The information in this presentation represents a snapshot of work in progress within DMTF
- This information is subject to change without notice. The standard specifications remain the normative reference for all information.
- For additional information, see the DMTF website.
- This information is a summary of the information that will appear in the specifications. See the specifications for further details.

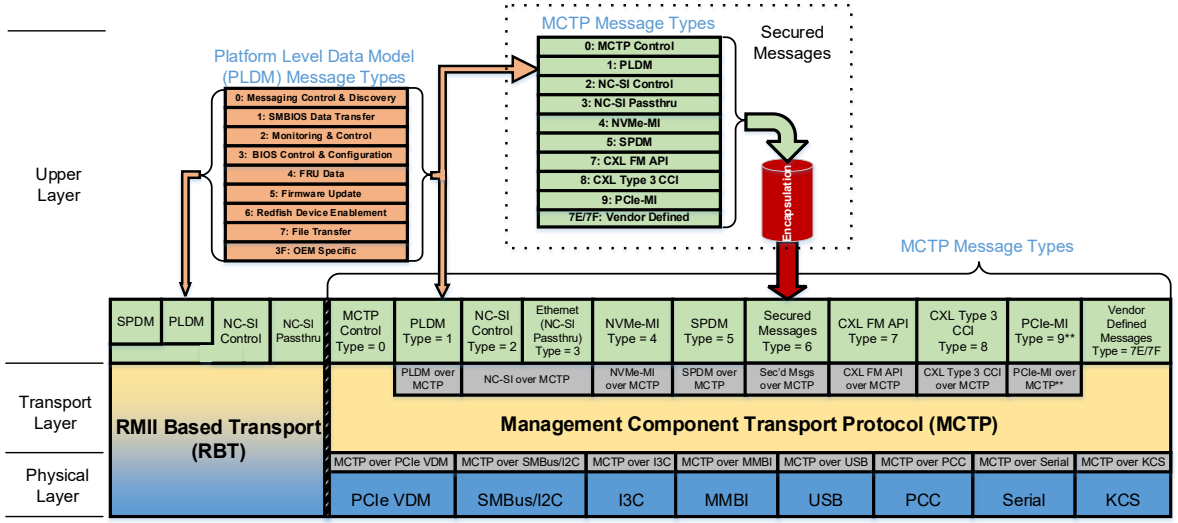
Outline

- Introduction
- MCTP Enhancements
- Simple MCTP Implementation
- PLDM Enhancements
- NC-SI over Ethernet over USB
- PMCI Tools/Libraries
- Call to Action

DMTF PMCI Working Group

- DMTF creates open manageability standards
- Platform Management Communications Infrastructure (PMCI) Working Group
  - Provides “Inside the box” (on-platform) communication and functional interfaces
  - Covers components within the platform management subsystem
  - Creates specs for MCTP, PLDM, NC-SI & FRU
- Applicability to OCP: platform components communication, device management





** Work in Progress

Process Improvements for Specifications

- DMTF has moved to GitHub based lightweight and fast specification development
- Multiple authors simultaneously contribute and author a specification
- Pull requests and issues can be reviewed in the same meeting
- PMCI WG has > 80% specs in GitHub development now
- PMCI WG plans to have ~100% of spec development on GitHub by 2027

MCTP Enhancements

- DSP0236: Management Component Transport Protocol (MCTP) Base Specification
  - Discovery related
    - Control command for endpoint advanced capability discovery
    - Updated discovery flow for hot-inserted devices
    - “Prepare for Endpoint Discovery command” requirements clarifications
  - Support for a single device with multiple Endpoint IDs (EIDs)
  - Support for Negotiated Transmission Unit (NTU)
  - Resiliency and recovery features and associated commands
    - Query Supported Recovery Actions
    - Recovery Request
    - Query MCTP Device Type
  - Commands to request and release EID pool – enabling non-consecutive EIDs
- DSP0238: MCTP over PCIe VDM – added support for PCIe Gen7

Released in Sept.

Simple MCTP Implementation

- MCTP is simple to implement as proven by universal acceptance and deployment
- What further simplifies MCTP for a given environment?
  - **Simple addressing** schemes
    - **Physical addressing** → no need to use logical addressing (EIDs) **or**
    - **Fixed physical topology** → static EID assignment are fine, no dynamic discovery
  - All endpoints on a single physical network → **No MCTP bridging or routing** needed
  - Every upper layer message fits in a single MCTP packet → **No message reassembly**
  - QoS is not required → **No** need to implement **rate limits**
  - Baseline Transmission Unit (BTU) of fixed MTU is sufficient → **No MTU negotiation** is required
  - Application-level recovery for MCTP communications → **No device level recovery** needed
  - OCP recovery is unaffected by simple MCTP
- DMTF plans to publish a specification to guide simple MCTP implementations

PLDM Enhancements

- DSP0240 – Platform Level Data Model (PLDM) Base Specification v1.3
- DSP0248 – Platform Monitoring and Control v1.4
- DSP0251 - Platform Level Descriptor (PDR) is now as a separate specification

Scalable Sensors, Structured Attributes, Clone PDRs
Available Now

- DSP0242 – PLDM for File Transfer v1.1

New Feature – Ability to Write files Coming Soon

- DSP2070 – PLDM Host SoC Modeling
  - PLDM model for SoC covering compute, memory, PCIe, and integrated NICs
- DSP2091 – PLDM Bulk Telemetry Modeling
  - Provides both a ‘push’ and ‘pull’ model to transfer bulk telemetry data
- DSP2092 – PLDM Crash Log Modeling
  - Collection of data when an entity (device) transitions to an error state
- DSP2093 – PLDM Watchdog Timer Modeling
  - Defines entity types, state/numeric sensors, and effecters for watchdog timer
- DSP2094 - RAS API PLDM Modeling
  - Enables interoperability of RAS API Endpoints (EP) and Initiators to communicate over PLDM

New Use Cases w/ PLDM Models
In Process

USB as a Sideband Interface

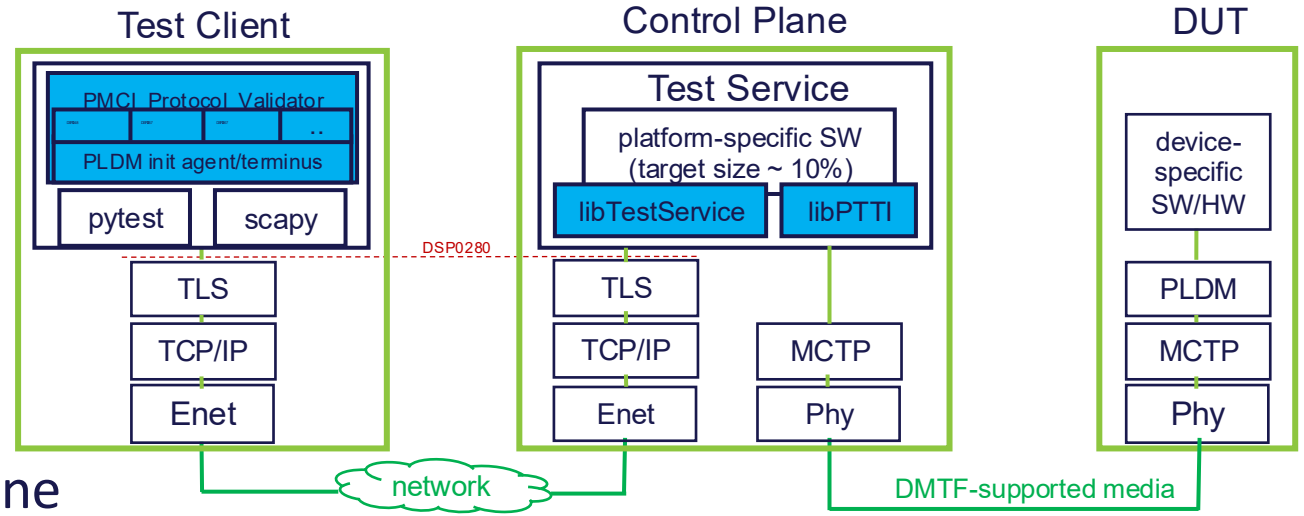
- OCP NIC specification already includes USB interface
- PCI CEM 6.x is adding USB signals for sideband management
- USB Adoption in OCP
  - OCP NIC
  - OCP DC-MHS/M-XIO
  - OCP DC-SCM
  - OCP OAI/UBB/OAM
  - OCP recovery
  - OCP OBMF-ICP
- DSP0283: MCTP over USB enhancements (packet splitting & resiliency) in 1.1.0

DSP0296: NC-SI over Ethernet over USB Binding

- Preserves USB Ethernet NIC model (common in BMCs)
- Leverages Ethernet over USB specification
- Uses Ethernet for NC-SI Pass-Through and NC-SI control (same as in DSP0222)
- Avoids an additional encapsulation layer compared to NC-SI/MCTP/USB
- Supports multiple NC-SI packages on a single USB bus
- Supports Ethernet Control Model (ECM) and Network Control Model (NCM)
- Allows co-existence with MCTP over USB traffic (for PLDM, SPDM, etc.)
- Already published specification

PMCI Open-Source Tools

- Protocol validation
- DUT independence
- Leverages pytest/scapy
- Test Service
  - Part of the Target (e.g., BMC) Control Plane
  - Goal is to achieve 80 - 95% common code
  - Defined Test Service / Control API to encourage “Accept As Is” adoption
- Test Client (open source)
  - PMCI Protocol Validator (Python with Scapy Library)
  - PMCI WG Protocols: PLDM Monitor & Control, NC-SI
  - PMCI Test Tools Interface and Design Specification (DSP0280) compliant implementation



Single Open-Source Tool that people can contribute to

PMCI Tools: Accomplishments and Future

- 2026
  - DSP0280 PMCI Test Tools Interface and Design Specification v1.1.0 is published
  - Open-Source Test Service Code is public
  - Open-Source Test Client Code is public including base PLDM and NC-SI functionality
    - Modular design allows open contributions for other upper layers over MCTP
- 2027-Onwards (Possible Future)
  - Increase the number of test cases in the protocol-validator (Test Client)
  - Add automation examples to exercise the test cases
  - Encourage adoption of lib-test-service (Test Service) for control planes such as OpenBMC
  - Encourage additional participation in the PMCI Tools TF to further development

Call to Action

- Contribute to these next generation specifications for OCP use cases
  - Direct through the PMCI Workgroup Group of DMTF or
  - DMTF Feedback Portal -> https://www.dmtf.org/standards/feedback
- Submit proposals for specification considerations with OCP projects/platforms
- Ensure manageability within OCP take advantage of MCTP/PLDM/NC-SI capabilities
- Contribute to PMCI tools repository link?
- Additional resources:
  - PMCI: https://www.dmtf.org/standards/pmci
  - PMCI Forum: https://pmciforum.com

Thank You!

